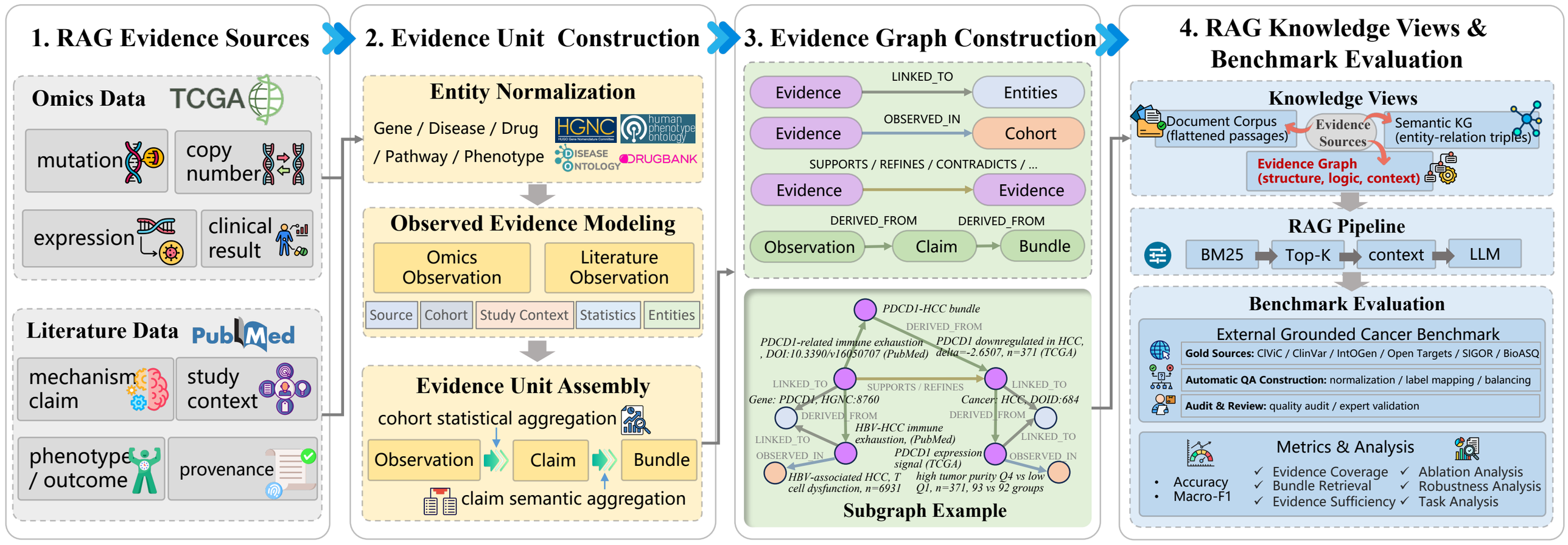

1. RAG Evidence Sources
Omics Data
TCGA
mutation
copy number
expression
clinical result
Literature Data
PubMed
mechanism claim
study context
phenotype / outcome
provenance
2. Evidence Unit Construction
Entity Normalization
Gene / Disease / Drug / Pathway / Phenotype
HGNC
human phenotype ontology
DISEASE ONTOLOGY
DRUGBANK
Observed Evidence Modeling
Omics Observation
Literature Observation
Source
Cohort
Study Context
Statistics
Entities
Evidence Unit Assembly
cohort statistical aggregation
Observation
Claim
Bundle
claim semantic aggregation
3. Evidence Graph Construction
Evidence
LINKED_TO
Entities
Evidence
OBSERVED_IN
Cohort
SUPPORTS / REFINES / CONTRADICTS / ...
Evidence
Evidence
DERIVED_FROM
DERIVED_FROM
Observation
Claim
Bundle
PDCD1-HCC bundle
DERIVED_FROM
PDCD1-related immune exhaustion , DOI:10.3390/v16050707 (PubMed)
PDCD1 downregulated in HCC, delta=-2.6507, n=371 (TCGA)
LINKED_TO
SUPPORTS / REFINES
LINKED_TO
Gene: PDCD1, HGNC:8760
Cancer: HCC, DOID:684
DERIVED_FROM
DERIVED_FROM
HBV-HCC immune exhaustion, (PubMed)
LINKED_TO
LINKED_TO
OBSERVED_IN
PDCD1 expression signal (TCGA)
OBSERVED_IN
HBV-associated HCC, T cell dysfunction, n=6931
high tumor purity Q4 vs low Q1, n=371, 93 vs 92 groups
Subgraph Example
4. RAG Knowledge Views & Benchmark Evaluation
Knowledge Views
Document Corpus (flattened passages)
Evidence Sources
Semantic KG (entity-relation triples)
Evidence Graph (structure, logic, context)
RAG Pipeline
BM25
Top-K
context
LLM
Benchmark Evaluation
External Grounded Cancer Benchmark
Gold Sources: ClViC / ClinVar / IntOGen / Open Targets / SIGOR / BioASQ
Automatic QA Construction: normalization / label mapping / balancing
Audit & Review: quality audit / expert validation
Metrics & Analysis
Accuracy
Macro-F1
✓ Evidence Coverage
✓ Bundle Retrieval
✓ Evidence Sufficiency
✓ Ablation Analysis
✓ Robustness Analysis
✓ Task Analysis